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CardiaTics: An explainable AI integrated heart disease diagnosis model with feature engineering and stacked ensemble approach

Partho Ghose¹ and Hasan Jamil^{2*}

*Correspondence:

Hasan Jamil

jamil@uidaho.edu

¹Department of Biological and Agricultural Engineering, Texas A&M University, College Station 77843, Texas, United States

²Department of Computer Science, University of Idaho, 875 Perimeter Drive, Moscow 83844, Idaho, United States

Abstract

Heart disease is a leading global cause of morbidity and mortality. Accurate and prompt diagnoses are crucial for its effective prevention and management. Integrating multiple machine learning algorithms, this research introduces a stacked ensemble machine learning model, called CardiaTics (stands for Cardiac DiagnosTics), toward improving heart disease detection. We detect outliers and remove them as a first-step to ensure data quality and maintain integrity. Ten distinct machine learning algorithms are then individually applied, culminating in the creation of a stacked ensemble model. We use feature engineering to refine the model further applying three well-known techniques –Pearson correlation, Chi-Square Test (Chi-2), and Recursive Feature Elimination. The implementation of these techniques on the benchmark dataset results in an optimized feature set. Experimental results show that CardiaTics delivers 89.3% accuracy on raw data, and significantly improves its accuracy after feature selection to 93.3%, outperforming the individual classifiers. However, can human professionals rely on algorithms for prediction when the underlying process is not fully understood? To address concerns regarding interpretability, trust, and transparency in black-box predictions, we propose utilizing SHapley Additive exPlanations (SHAP) and Explain Like I'm 5 (ELI5) in the second phase to elucidate feature importance in our model. The SHAP summary plots of CardiaTics reveal that the positive and negative contributors to heart disease are comparable, thereby enhancing the model's interpretability and reliability and helping refine the decision-making process.

Keywords Heart disease, Machine Learning, Ensemble Learning, Feature Engineering, Explainable AI, SHAP

Introduction

Cardiovascular diseases (CVDs) pose a significant global health threat. American Heart Association reports it claims about 17.9 million lives globally each year [1]. In the United States alone, a stroke-related fatalities occurs every three minutes and is considered as one of the leading causes of death [2, 3]. High rates of stroke-related fatalities are also observed in regions such as Central, Southeast, and East Asia, Oceania, and sub-Saharan

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Africa [4]. Even in the cases of non-fatal strokes, serious heart complication often ensue post stroke requiring significant care regimes [5].

Traditional diagnostics of heart diseases rely on archaic protocols involving patient medical history, physical examinations, and presenting symptoms [6] that are plagued with potential inaccuracies, delays, and increased costs due to human errors [7]. Recent emergence of computational methods have shown promise in addressing these shortcomings potentially reducing mortalities and healthcare costs. These computational models are also helping in early detection and in improving patient outcomes [8]. In particular, heart disease prediction using machine learning (ML) has emerged as a leading area of research and investigation [9]. Unlike conventional methods, comprehensive ML based methods leverage a multitude of health variables to predict patients' likelihood of having CVD [7].

Among these ML approaches, an approach called ensembling [10] has gained prominence in the area of heart disease diagnoses and predictions. Recently, several ensembling adaptations were also shown to have been instrumental in improved and effective prediction toward CVD identification [11, 12]. This technique combine multiple models for enhanced performance. In this paper, our goal is to propose a novel adaptation of the ensembling model combined with effective feature selection techniques for cardiac disease prediction and diagnosis [11].

Though ML based diagnoses approaches are powerful and promising, they also have wide ranging significant limitations. Among them, feature selection and engineering have attracted significant attention from the research community due to the roles they play in the prediction models, ensembling in particular.

In the context of heart disease prediction [13, 14], various studies have examined the impact of specific features, including ST depression, ST slope, maximum heart rate achieved, and cholesterol levels [15, 16]. These investigations provide valuable insights into the significance of these parameters for cardiovascular health assessments.

Feature engineering is crucial for enhancing the performance of machine learning models, particularly when dealing with complex medical datasets. Previous research [17, 18] has highlighted the importance of selecting and transforming relevant features to achieve accurate disease prediction. Techniques such as dimensionality reduction [19, 20], feature scaling, and identification of critical biomarkers [21–23] significantly contribute to the optimization of diagnostic models.

Like most other methods, feature selection play a critical role in refining model inputs in ensembling. Existing methods [8, 24] struggle to identify and utilize the most relevant features for heart disease prediction. For instances, Pearson correlation and Chi-Square Test may overlook subtle interactions and fail to adapt to the dynamic nature of cardiovascular dynamics [8]. Therefore a compensatory assembly of multiple feature selection technique to help select the most significant features that potentially contribute to accurate predictions of heart diseases deserves due attention.

In this paper, we employ three feature selection strategies such as Pearson correlation, Chi-Square Test (Chi-2), and Recursive Feature Elimination (RFE)) to select the most effective features so that the model can focus only on the relevant features to learn. We apply three classifiers – LR, RF and Light Gradient-Boosting Machine (LightGBM) to observe the feature contribution of these methods. Later, we combine those features which are used by all six methods for prediction and contribute with high scores. Finally,

we incorporate model interpretability tools, SHapley Additive exPlanations, more commonly known as SHAP values, to support explainability toward improving our understanding of the intricacies of heart disease dynamics. This strategic approach seeks to overcome the challenges associated with the “black-box” nature of some ML models and aims to provide more reliable and interpretable predictions for improved clinical decision-making. The proposed CVD prediction system CardiaTics makes several interesting contributions as summarized below.

1. With a view to enhance model’s prediction, we apply three feature selection techniques such as Pearson correlation, Chi-2 test, and Recursive Feature Elimination (RFE)) to select the most influential factors contributing to heart disease.
2. To understand the quality and effectiveness of CardiaTics, we use several evaluation metrics to measure performance, e.g., accuracy, precision, sensitivity, specificity, F-1 score, Logloss, Matthew correlation coefficient (MCC) and AUC-ROC for reliable prediction of heart diseases.
3. We also incorporate model interpretability tools such as ELI5 and SHAP values in CardiaTics for a deeper understanding of heart disease dynamics with a quest to offer clarity in interpreting ML models’ outcomes.

The presentation of the remainder of the article is planned as follows. We discuss the methodology of the CardiaTics system in Sec “[Methodology](#)” and present the foundation of model building based on our feature selection technique and the explainability of the model, and its evaluation metrics. Experimental evaluation follows in Sec “[Experiments, results and discussions](#)” before we conclude in Sec “[Conclusion](#)”

Related research

Though ML-based diagnostic approaches are powerful and promising, they also have wide-ranging significant limitations. Among them, feature selection and engineering have attracted significant attention from the research community due to the roles they play in the prediction models, ensembling in particular.

Stacked ensemble learning models have emerged as a powerful medical diagnosis technique because they enhance predictive performance using diverse algorithms. Stacking [25] combines strengths from multiple base models, compensating for individual weaknesses and improving overall accuracy. Previous health research [24, 26], including heart disease diagnosis, attests to the effectiveness of ensemble learning and showcases enhanced sensitivity and specificity. In the heart disease prediction spectrum, studies [12, 27] have individually examined various base classifiers such as Random Forest (RF), Support Vector Machines (SVM), and Gradient Boosting Machines (GDM) [8, 9]. They highlight the pivotal role of selecting and combining base models toward significantly influencing the stacked ensemble’s predictive capabilities [28].

One of the voting ensemble methods [29] incorporates multiple machine learning techniques, yielding an accuracy of 90%, albeit with a poor overall performance. One of the hybrid intelligent machine-learning-based predictive approaches for heart disease detection [10], on the other hand, utilized data from Cleveland Clinic’s heart disease cases and integrated seven well-known classifiers with three feature selection algorithms to achieve 89% accuracy using Logistic Regression (LR). Another intelligent healthcare

monitoring system for heart disease prediction used a deep learning-based ensemble method to achieve promising results [30].

The literature highlights the critical importance of selecting and combining base models to enhance the predictive capabilities of stacked ensemble models [28]. A thorough understanding of the strengths and limitations of individual classifiers is essential for building a robust stacked ensemble.

Abdar et al. [31] made a significant contribution by introducing a novel machine learning technique for detecting coronary artery disease. They used an N2Genetic optimizer, derived from Genetic Algorithm (GA) and Particle Swarm Optimization (PSO), for data processing and feature selection, achieving an accuracy of 93.08% with the N2Genetic-nuSVM approach. This method, however, relied solely on GA and PSO techniques, which are known to be time-consuming due to the elimination of redundant features.

Rahma et al. [29] proposed a voting ensemble method that incorporated multiple machine learning techniques, achieving an accuracy of 90%. However, their study lacked any discussion of feature engineering techniques, and the overall performance was relatively poor.

Shah et al. [10] introduced a hybrid intelligent machine-learning-based predictive approach for heart disease detection. Using data from Cleveland heart disease cases, their system integrated seven well-known classifiers with three feature selection algorithms, achieving the highest accuracy of 89% with logistic regression.

Ali et al. [30] presented an intelligent healthcare monitoring system for heart disease prediction using a deep learning-based ensemble method, achieving promising results. Despite these advances, there is still a need for improved performance evaluation of these diagnostic models. Many existing models offer only limited performance measurements and do not adequately address the contribution of each feature to the detection of heart disease.

In line with these efforts, several recent studies also have introduced advanced ensemble-based frameworks for heart disease prediction. MU Rehman et al. [32] proposed a hybrid machine learning approach combining Particle Swarm Optimization (PSO) with an Artificial Neural Network (ANN) to enhance the prediction of coronary heart disease. Their framework incorporated mutual information (MI) for feature selection and applied the Synthetic Minority Oversampling Technique (SMOTE) to handle class imbalance in the dataset. They achieved a prediction accuracy of 97% using the PSO-ANN model, outperforming conventional classifiers like Logistic Regression and Random Forest, which reached up to 95.8% accuracy on the NHANES dataset. However, the approach had notable limitations, including high computational cost due to the iterative nature of PSO and a lack of external validation, which raises concerns about the model's generalizability to unseen clinical datasets. Similarly, M Darshan Teja and G Mokesh Rayalu [33] presented a hybrid voting ensemble combining LR, RF, and MLP, which achieved 93.1% accuracy, although their manual hyperparameter tuning and lack of external validation limited scalability. M.T. Ashfaq [34] introduced a deep learning model, FCW-Net, for heart disease prediction, combining ProWSyn oversampling to address class imbalance and PCA for dimensionality reduction. The model achieved strong results with an AUC-ROC of 0.9622, accuracy of 92.37%, and recall of 97.21%. To enhance transparency, SHAP analysis was used to interpret feature contributions. However, the model's complexity increases computational cost, and despite SHAP, it remains

partially black-box, limiting full clinical interpretability. In [35], the authors proposed a stacked ensemble model for heart disease detection, incorporating outlier removal and feature selection to enhance performance. The model achieved 93.3% accuracy. While SHAP was used for global interpretability, the study lacked local explainability, limiting insights into individual predictions.

Table 1 summarizes the key contributions and limitations from recent research on heart disease prediction. Current research is actively addressing these gaps, focusing on refining and balancing assessment metrics to increase diagnostic accuracy for this critical disease.

Methodology

This research aims to develop an automated system for detecting heart disease, which could benefit physicians and patients. The subsequent sections provide a brief overview of the materials and methods used to achieve this objective. The research methodology comprises five significant steps, illustrated in figure 1.

Data description

The Heart Disease data used in this study is a synthesis of data sourced from the UCI Machine Learning Repository [37] and contains 11 features that can be used to predict the presence of heart disease as summarized in Table 2. The target variable is a binary attribute indicating the presence or absence of heart disease. The figure 2 shows the distribution of heart disease cases in the dataset. The pie chart indicates that 53% of individuals have heart disease, while 47% are normal and the bar chart confirms this with 629 heart disease cases and 561 normal cases.

Table 1 Summary of recent works on heart disease prediction using ensemble models

Reference	Method	Feature Selection	ACC (%)	Limitations
[31]	N2Genetic-nuSVM	GA, PSO	93.08	Time-consuming feature elimination
[36]	SVM, KNN, MLP	ANOVA, EDA, PCC	72	Lack of feature engineering, relatively poor performance
[10]	Hybrid Predictive Approach	Multiple algorithms	89	Limited feature selection techniques
[30]	Deep Learning Ensemble	-	-	Needs enhanced performance evaluation
[24]	KNN, SVM, LR	LASSO, ANOVA, MultiSURF	97.5	Limited analysis on model impact
[18]	RNN+LSTM	Relational Approach	91	No detailed feature contribution analysis
[8]	GBM, SVM, KNN, Bagging	One-Hot, SMOTE	91.8–92	Focus on individual classifiers; lacks feature attribution
[9]	XGBoost classifier	Undersampling	87	No detailed feature contribution analysis
[32]	PSO-ANN Hybrid	MI, SMOTE	97	High computational cost; no external validation
[33]	Voting Ensemble (LR, RF, MLP)	Manual selection	93.1	Manual tuning; no external validation
[34]	FCW-Net (Deep Learning)	PCA, ProWSyn	92.37	High complexity; limited interpretability despite SHAP
[35]	Stacked Ensemble	Pearson, Chi2, RFE	93.3	No local interpretability; limited per-instance insights

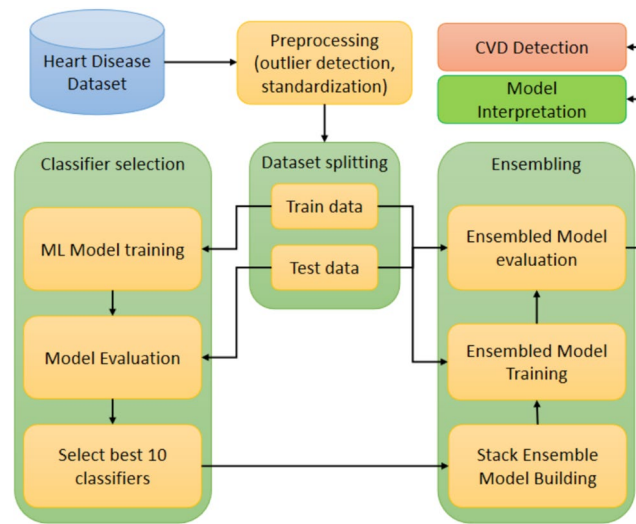


Fig. 1 Framework of heart disease detection model

Table 2 Descriptions of the dataset

No.	Name of the features	Values
1.	Age	30~77
2.	Sex	The patient's gender, male or female
3.	Chest Pain Type	The type of chest pain experienced by the patient: i. TA: Typical Angina ii. ATA: Atypical Angina iii. NAP: Non-Anginal Pain iv. ASY: Asymptomatic
4.	RestingBP: The patient's resting blood pressure (mmHg)	94–200 mm Hg
5.	Cholesterol: The patient's serum cholesterol (mg/dl)	120–564 mg/dl
6.	FastingBS: The patient's fasting blood sugar	i. 1 if glucose > 120 mg/dl ii. 0 otherwise
7.	RestingECG: Resting electrocardiogram results	i. Normal ii. ST: having ST-T wave abnormality (T wave inversions and/or ST elevation or depression of > 0.05 mV) iii. LVH: showing probable or definite left ventricular hypertrophy by Estes' criteria
8.	MaxHR: Maximum heart rate achieved, beats per minute	71–202
9.	Exercise-induced angina	1 = yes, 0 = no
10.	Old peak= ST depression induced by exercise relative to rest	0–6.2.2
11.	Slop of the peak exercise ST segment	0= normal, 1= up sloping, 2 = flat, 3 = down sloping
12.	Target	0= the patient is healthy, 1= the patient is suffering from heart risk

Outlier detection and data normalization

An outlier is a data point that significantly deviates from the typical values within a dataset, being either exceptionally large or exceptionally small. It could indicate a potential data entry error or represent legitimate but exceptional data. Outliers can be detected in different ways. Here, we used Z-score technique as depicted in Eq. (1). Here, x is the score, μ is the mean, and σ is the standard deviation.

$$Z = \frac{x - \mu}{\sigma} \quad (1)$$

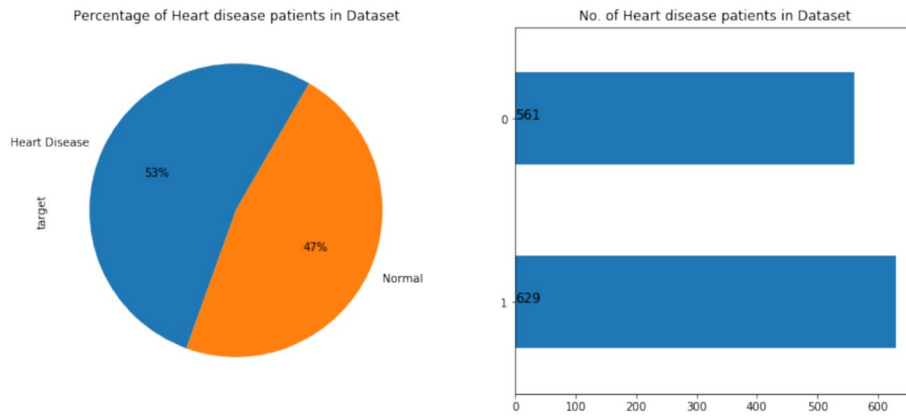


Fig. 2 Distribution of heart disease and normal cases in the dataset

Setting a threshold for filtering, we only plan to keep the data that has $z < 3$, and remove the outliers. However, in our data set no outliers were found.

To normalize the data, we first encode categorical variables as dummy variables and separate the feature and the target variable. Afterward, we normalize all the numeric features in the range of 0 to 1 using min-max feature scaling transformation. The original data are transformed linearly as part of the normalization process to generate a balance of value comparisons between the data before and after the operation. This process can be defined as Eq. 2.

$$\rho_{\text{new}} = \frac{\rho - \min(\rho)}{\max(\rho) - \min(\rho)} \quad (2)$$

where ρ_{new} = normalized data, ρ = old data, $\max(\rho)$ = maximum value of old data, $\min(\rho)$ = minimum value of old data.

Feature selection techniques

We follow a systematic exploration of feature selection techniques to enhance the robustness and effectiveness of our model. We employed three distinct methods, namely Pearson correlation, Chi-Square Test (χ^2), and Recursive Feature Elimination (RFE). Pearson correlation coefficient measures the linear correlation between two variables. It is commonly used in feature selection to identify features that have a strong linear relationship with the target variable. The formula for Pearson correlation coefficient (often denoted as r) is as shown in Eq. 3.

$$r = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum (X_i - \bar{X})^2 \sum (Y_i - \bar{Y})^2}} \quad (3)$$

where, r is the Pearson correlation coefficient, X_i and Y_i are the individual data points, $\bar{X}$ and $\bar{Y}$ are the means of the variables X and Y , respectively.

Chi-2 measures the independence between variables. It is often used in feature selection to select the most relevant features for classification tasks as shown in Eq. 4.

$$\text{Chi} - 2 = \sum \frac{(O_i - E_i)^2}{E_i} \quad (4)$$

where, O_i is the observed frequency for category i , and E_i is the expected frequency for category i .

RFE is a feature selection method that recursively removes features, fitting the model and evaluating its performance until the desired number of features is reached. The utilization of these techniques is pivotal in identifying and retaining the most relevant features that significantly contribute to the predictive power of our model. Pearson correlation gauges linear relationships between variables, the Chi-2 test assesses the independence of categorical features, and the RFE systematically eliminates less informative features. Although deep feature learning may improve performance, we employ lightweight and interpretable feature selection to maintain transparency and clinical relevance. These feature selection steps ensure that our model is equipped with the most discriminative and informative attributes, thereby optimizing its performance and interpretability.

Model building

We now focus on the process of model construction, beginning with the cross-validation and baseline model selection for the proposed framework. Through systematic evaluation and comparison, we identify the most effective models to serve as the foundation for further refinement. Finally, we introduce the proposed stacked ensemble model that combines the strengths of each of its component models for an improved predictive accuracy and robustness compared individual models alone.

Cross-validation and model selection

We select various baseline models and conduct 10-fold cross-validation to identify the most effective baseline algorithms based on their accuracy. These selected models are then used as the base layer in the stacked ensemble method. The 10-fold cross-validation for each model is determined using Eq. 5.

$$cva_k = \frac{1}{k} \sum_{i=1}^k Accuracy_i \quad (5)$$

where, k is the number of folds, cva_k denotes the accuracy of a model employing the k -fold cross-validation. The top ten well performing base models selection process can be defined as the Eq. 6 below.

$$\forall \{x \in N, 0 < x < 10\}, model_x = O_{max...min}(cva_1, cva_2, \dots, cva_n) \quad (6)$$

where, $model_x$ = selected models, and $O_{max...min}$ = a function that arranges cross-validation accuracies of n different models in descending order.

Proposed model

Stacking, also known as super learning, represents a category of algorithms that involves training a second-level "meta learner" to optimize the combination of diverse base learners [38]. Unlike bagging and boosting, stacking aims to combine robust and varied sets of learners. We selected stacking rather than bagging/boosting because (i) we combine heterogeneous learners whose errors are complementary, and (ii) a meta-learner can learn nonlinear interactions among base predictions. Bagging is most effective when

aggregating many instances of the same high-variance learner, while boosting focuses on sequential error correction; both are valuable baselines, but stacking is better suited to fusing diverse model families.

Based on the cross-validation accuracy performance of different baseline models, we select the top ten performing models for the base level (level 0) of the stacked ensemble. This ensemble is designed to outperform individual ML models. In our study, we designate Multilayer Perceptron (MLP) as our super learner or generalizer. The workflow of the 10-fold stacking model is presented in Fig. 3. The process proceeds as follows.

The training set is divided into ten initial segments, nine of which serve as training data for individual classifiers, and any part of each fold acts as the test data. The fundamental classifiers include RF, MLP, K-Nearest Neighbors (KNN), Extra Tree Classifier (ETC), eXtreme Gradient Boosting (XGB), Support Vector Classifier (SVC), Singular Value Decomposition (SGD), Adaboost Classifier (ABC), Classification And Regression Tree (CART), and Gradient Boosting Machine (GBM). Each classifier is trained and evaluated over the randomly selected segment in each of the ten folds. Consequently, each classifier generates ten sets of test results, and their union constitutes the entire training set.

To identify the optimal ensemble configuration, we employ a Multi-Layer Perceptron (MLP) as the meta-learner to integrate the predictions of the base classifiers. MLP was chosen due to its ability to model complex, non-linear relationships among the outputs of heterogeneous base learners, which enhances its capacity to learn intricate decision boundaries at the meta-level [39]. Compared to linear models such as logistic regression or simpler learners like decision trees, MLP offers superior representational power, enabling it to exploit subtle inter-model dependencies. Furthermore, its scalability and robustness to overfitting, when combined with appropriate regularization and early stopping, make it a suitable choice for aggregating diverse model outputs in high-dimensional decision spaces [39].

This ensemble technique takes advantage of the collective wisdom of multiple base learners, each trained using k -fold (in our case $k = 10$) cross-validation. In this framework, the ensemble's prediction, denoted as $\hat{f}_{CardiaTics}(x)$, is computed by combining the predictions of individual base learners through a weighted averaging scheme as expressed in Eq. 7.



Fig. 3 Procedure of proposed ensemble learning

$$\hat{f}_{CardiaTics}(x) = MLP \left(\frac{1}{N} \sum_{i=1}^N \left(\frac{1}{k} \sum_{j=1}^k w_{i,j} \times \hat{y}_{i,j} \right) \right) \quad (7)$$

Here, N represents the number of base learners, k signifies the number of folds in cross-validation, $\hat{y}_{i,j}$ denotes the prediction of the i -th base learner on the j -th fold. The w_{ij} denotes the weight of the connection between neuron i in the input layer and neuron j in the output layer. The MLP meta learner synthesizes these weighted predictions to yield the final ensemble output. By leveraging the strengths of diverse base learners and incorporating the MLP's capacity for nonlinear modeling, this stack ensemble methodology excels at tackling complex predictive tasks, offering superior generalization performance and robustness in real-world scenarios.

XAI

Explainable Artificial Intelligence (XAI) has emerged as a vital area of research aimed at making complex machine-learning models more transparent, interpretable, and trustworthy. The core objective of XAI is to bridge the gap between highly accurate but opaque "black-box" models and the growing demand for accountability, especially in critical domains, such as healthcare. By providing human-understandable insights into how models arrive at their predictions, XAI enables domain experts and stakeholders to make informed decisions, validate model behavior, and build trust in AI-driven systems.

The importance of explainability has grown considerably in recent years, not only from a scientific and ethical standpoint but also due to evolving legal and regulatory frameworks. Unlike a decade ago, when automated algorithms were often deployed without questioning their internal decision logic, current applications increasingly require transparency by design. This shift is particularly evident in the European Union, where new regulations mandate the traceability and interpretability of AI systems, especially those used in high-stakes areas, such as medicine, finance, and law. These regulatory developments reflect a broader global trend toward the responsible and auditable deployment of AI technologies.

The need for explainable AI is even more pronounced in healthcare. Although AI holds immense potential to enhance diagnostics, treatment planning, and patient outcomes, its adoption in clinical practice remains limited [40]. Although the lack of labeled medical data is often cited as a barrier, a recent work by Pagallo et al. highlights that regulatory uncertainty and legal constraints are equally significant factors hindering the integration of AI into healthcare workflows. Without clear frameworks that balance innovation with ethical oversight, the healthcare sector risks underutilizing AI and incurring opportunity costs in terms of efficiency, accuracy, and patient-care outcomes [41].

In response to these challenges, a wide array of XAI techniques have been developed, making it one of the most dynamic and rapidly evolving in AI research. These techniques are generally categorized into global and local [42]. Global methods aim to explain the overall logic and behavior of a model, helping users understand how different input features influence predictions. By contrast, local methods focus on individual predictions and offer explanations tailored to specific instances. Both approaches are essential to ensure transparency and trust, particularly when AI systems are used in sensitive and regulated environments.

However, the proliferation of XAI methods presents a new challenge for practitioners: how to choose the most suitable technique for their specific use case. As XAI matures, there is growing recognition of the gap between theoretical advancements and real-world implementation. To address this, Retzlaff et al. [43] proposed a comprehensive decision-making framework to guide data scientists and engineers in selecting between ante-hoc (inherently interpretable) and post hoc (retrospectively interpretable) approaches. Their work provides actionable design guidelines that align the XAI method selection with application requirements, model complexity, and end-user needs [43]. This framework is particularly valuable for practitioners working in domains such as healthcare, where explainability is not only desirable, but also often mandated by policy [44].

Above all, as AI continues to permeate decision-critical sectors, explainability is no longer optional; it is a prerequisite. Understanding both the technical spectrum of XAI methods and the broader legal and ethical contexts is essential for deploying AI responsibly and effectively.

Various interpretability techniques are available, each with distinct advantages and limitations based on the type of model for which they are designed. Local interpretable model-agnostic explanation (LIME) is a widely used method that generates explanations for individual predictions by approximating a model's decision boundary with a simpler, interpretable surrogate model. Although effective, LIME can lead to unstable explanations owing to its reliance on local approximations [45]. This can become problematic in clinical environments, where consistent and repeatable results are essential for making informed decisions. Grad-CAM (Gradient-weighted Class Activation Mapping) is another popular method, particularly in image-based applications, such as medical imaging. It visualizes the important regions in an image that contribute to a model's decision. However, Grad-CAM is limited to convolutional neural networks (CNNs), which makes it unsuitable for models beyond CNNs, such as those used in medical AI applications that combine different neural architectures [45]. Layer-wise Relevance Propagation (LRP) works by propagating relevance scores through a network, offering insights into the features that influence model predictions. While it has been effective in certain medical imaging tasks, LRP is also architecture-dependent, meaning that it is not as flexible across diverse model types.

In contrast, Shapley Additive Explanations (SHAP) and ELI5 offer a more versatile and robust approach, especially for the various models we work with in medical AI. SHAP provides model-agnostic explanations by fairly distributing feature importance and ensuring consistency and transparency across both local and global interpretability [45]. This is crucial for healthcare applications, where understanding the contribution of each feature is important not only for accuracy but also for patient safety and clinical decision-making. ELI5, a complement to SHAP, provides a simplified and intuitive interface for visualizing feature contributions, making it accessible to clinicians and non-technical users. Together, these two methods overcome the limitations of LIME, Grad-CAM, and LRP, as they are adaptable to different model types and provide a higher level of consistency and transparency. This makes SHAP and ELI5 particularly suitable for our study, where explanations that are reliable, interpretable, and comprehensible in a medical context are required. These methods perfectly align with the needs of clinical

AI applications, where transparency and trust in the system are paramount for practical implementation and widespread adoption [45].

Model evaluation metrics

We largely adopt the standard metrics such as accuracy, precision, sensitivity and specificity for model evaluation as follows.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (8)$$

$$Sensitivity = \frac{TP}{TP + FN} \quad (9)$$

$$Specificity = \frac{TN}{TN + FP} \quad (10)$$

$$Precision = \frac{TP}{TP + FP} \quad (11)$$

Using the above metrics, we define the *F1 – Score* as follows.

$$F1 - Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \quad (12)$$

Finally, we calculate the MCC score as follows.

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP) \times (TP + FN) \times (TN + FP) \times (TN + FN)}} \quad (13)$$

The LogLoss value calculated as in Eq 14 indicates how close a prediction probability comes to the actual/corresponding true value.

$$LogLoss_i = -\frac{1}{N} \sum_{i=1}^N (A_i \times \ln P_i + (1 - A_i) \times \ln (1 - P_i)) \quad (14)$$

where i is the given observation or record, A denotes the actual or true value, and P denotes the probability of the forecast. Moreover, we use Area Under the Curve depicted as ROC-AUC curve for the performance evaluation. It is a probability curve that plots True Positive Rate (TPR) or Sensitivity against False Positive Rate (FPR).

Experiments, results and discussions

Our goal is to help identify individuals with cardiac issues using UCI datasets. The experiments we have conducted for the CardiaTics system using machine having an 8th generation Intel Core i7 processor running at up to 3.1 GHz and 16 GB of RAM under Windows.

Base models selection for cardiaTics

The base models in ensemble learning play a critical role in determining the predictive performance and generalization capability of the final ensemble and thus their appropriate selection is critical. Based on the performance results presented in Table 3, it is evident that RF Classifier (with the criterion set to 'entropy'), MLP, KNN (with $n = 9$), ET

Table 3 valuation of base models with hyperparameter configurations and rationale

Model	Hyperparameters	Accuracy	Loss	Rationale
LR_L2	penalty='l2', solver='lbfgs'	0.857081	0.026972	L2 penalty reduces multicollinearity; logistic regression suits binary classification.
LDA	solver='svd'	0.850686	0.027725	SVD solver performs well with large features; maximizes class separability.
KNN7	n_neighbors=7	0.848513	0.044004	Moderate neighborhood captures local structure while reducing noise.
KNN5	n_neighbors=5	0.850663	0.041023	Smaller k values are sensitive to local patterns, increasing flexibility.
KNN9	n_neighbors=9, weights='uniform', algorithm='kd_tree'	0.866678	0.033055	Nine neighbors offer a good bias-variance tradeoff; uniform weights simplify prediction; kd_tree accelerates search.
KNN11	n_neighbors=11	0.854930	0.034817	Larger k smooths decision boundary, improving stability.
CART	criterion='gini'	0.877408	0.043381	Gini criterion efficiently splits features; simple tree structure supports interpretability.
NB	var_smoothing=1e-9	0.750446	0.056369	Default smoothing improves robustness; fast and simple baseline model.
SVM Linear	C=1.0, loss='hinge', kernel='linear'	0.853889	0.030047	C=1.0 balances margin and misclassification; linear kernel is suitable for high-dimensional, linearly separable data.
SVM RBF	C=1.0, kernel='rbf'	0.849577	0.036938	RBF kernel captures nonlinear patterns; default C ensures generalization.
AB	n_estimators=50, learning_rate=1.0	0.865614	0.031921	50 estimators offer controlled boosting depth; learning rate of 1.0 ensures consistent weight adjustment.
GBM	n_estimators=100, learning_rate=0.1	0.901842	0.032180	100 estimators and 0.1 learning rate support incremental learning; effective for capturing complex relations.
RF_Ent100	n_estimators=100, criterion='entropy', max_depth=None	0.933848	0.020864	Entropy criterion promotes informative splits; 100 estimators balance complexity and runtime; no depth constraint allows detailed learning.
RF_Gini100	n_estimators=100, criterion='gini'	0.934923	0.015623	100 trees provide robustness; gini criterion promotes balanced splits.
ET100	n_estimators=100, criterion='mse', max_depth=None	0.926401	0.014077	Randomized splits via 'mse' reduce overfitting; 100 estimators support generalization; no depth limit ensures pattern capture.
ET500	n_estimators=500	0.925337	0.015983	Increased estimator count enhances ensemble stability and predictive performance.
MLP	hidden_layer_ sizes=(100,), activation='relu', solver='adam'	0.866655	0.033809	Single hidden layer with 100 neurons captures moderate complexity; ReLU aids efficient gradient flow; Adam ensures fast convergence.

Table 3 (continued)

Model	Hyperparameters	Accuracy	Loss	Rationale
SGD3000	max_iter=3000, loss='squared_hinge', penalty='l2'	0.810192	0.080261	3000 iterations allow convergence; squared hinge loss maximizes margin; L2 regularization controls model complexity.
XGB_2000	n_estimators=2000, max_depth=5, learning_rate=0.05	0.922123	0.028748	High estimator count captures complex patterns; depth and learning rate tuned to prevent overfitting.
XGB_500	n_estimators=500, max_depth=5, learning_rate=0.05	0.925315	0.023075	500 estimators provide capacity; depth of 5 controls complexity; learning rate of 0.05 ensures gradual weight updates.
XGB_100	n_estimators=100	0.895424	0.030502	Smaller number of trees allows faster training with moderate performance.
XGB_1000	n_estimators=1000	0.918932	0.027165	Intermediate complexity; balances capacity and generalization.
ET1000	n_estimators=1000	0.926401	0.014077	Large ensemble (1000 trees) increases variance reduction and robustness to overfitting.

Table 4 Model performances before feature selection is applied

Model	Accuracy	Precision	Sensitivity	Specificity	F1 Score	ROC	LogLoss	MCC
CardiaTics	0.893617	0.871212	0.934959	0.848214	0.901961	0.891587	3.674396	0.788341
RF	0.889362	0.864662	0.934959	0.839286	0.898438	0.887123	3.821373	0.780215
MLP	0.838298	0.819549	0.886179	0.785714	0.851562	0.835947	5.585075	0.677074
KNN	0.800000	0.787879	0.845528	0.750000	0.815686	0.797764	6.907851	0.599458
EXtra tree classifier	0.889362	0.870229	0.926829	0.848214	0.897638	0.887522	3.821369	0.779355
XGB	0.855319	0.856000	0.869919	0.839286	0.862903	0.854602	4.997161	0.709875
SVC	0.812766	0.788321	0.878049	0.741071	0.830769	0.809560	6.466933	0.627138
SGD	0.757447	0.779661	0.747967	0.767857	0.763485	0.757912	8.377579	0.515264
Ada-boost	0.817021	0.812500	0.845528	0.785714	0.828685	0.815621	6.319943	0.633084
CART	0.846809	0.871795	0.829268	0.866071	0.850000	0.847670	5.291098	0.694584
GBM	0.838298	0.829457	0.869919	0.803571	0.849206	0.836745	5.585068	0.675997

(with $n_estimators = 100$), XGB (with $n_estimators = 500$), SVC (with a kernel set to 'linear'), Stochastic Gradient Descent (SGD), ABC, CART, and GBM exhibit superior performance as classifiers. Their respective accuracies are 0.857081, 0.866655, 0.866678, 0.926401, 0.925315, 0.853889, 0.810192, 0.865614, 0.877408, and 0.901842. These models are found to be better performers than their own versions with different parameters.

Pre feature engineering model evaluation

We present an initial assessment of our model's performance before implementing feature engineering on the data to establish a baseline understanding of its effectiveness compared to conventional models. The performance of all the models on raw data is summarized in Table 4 in which CardiaTics emerges as the top performer (marked in bold). In this table, CardiaTics has the highest test accuracy (0.894), sensitivity (0.935), specificity (0.848), and F1-score (0.902), and have the lowest LogLoss (3.674) and highest

ROC value (0.892). Conversely, the SGD classifier exhibits the poorest performance, with the lowest test accuracy (0.757), sensitivity (0.747), specificity (0.767), and F1-score (0.757), along with the highest LogLoss (8.377) and ROC value (0.75). The RF and ET Classifiers show similar and comparable performance in all metrics.

From Fig. 4, we can see the highest average AUC of 0.951 is gained by ET whereas the second-highest average AUC of 0.930 is obtained by RF classifier. CART has the lowest average AUC of 0.848. Notably, CardiaTics model has an AUC of 0.905, which is not the best but good enough.

Feature importance analysis

As alluded to before, we used three feature engineering techniques (Pearson correlation, Chi-2, and RFE) to identify the most relevant features. Once selected, we evaluate the behavior of these models post feature selection. Once again, we apply three classifiers - LR, RF, and LightGBM - to assess feature contributions.

The distribution of feature contributions across six methods is summarized in Table 5. Notably, all six methods identify st-depression as a significant contributor. Additionally, five methods identify st-slope-flat, max-heart-rate-achieved, exercise-induced-angina, cholesterol, and age as the next most contributing features. Four of the identify st-slope-upsloping, sex-male, chest-pain-type-non-anginal pain, and chest-pain-type-atypical angina as major contributors. However, st_depression is not identified as a significant contributor by most methods except RF and LightGBM. Thus, we remove this feature in subsequent model analyses.

Post feature engineering model evaluation

Once the most influential features are selected, we recalculate model performance to observe any improvement. Table 6 presents the model comparison post feature engineering step. As shown, CardiaTics remains the best model and the performance of

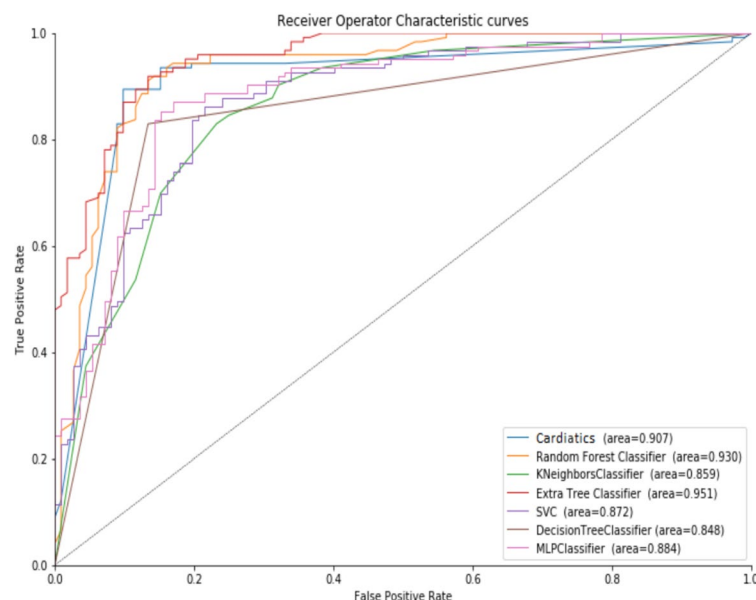


Fig. 4 ROC Curves Comparing Classification Models for Heart Disease Prediction with Corresponding AUC Scores. The Extra Tree Classifier achieved the highest AUC (0.951), followed by the Random Forest Classifier (0.930) and the custom CardiaTics ensemble model (0.907), indicating strong predictive performance.

Table 5 Feature selection through Pearson correlation, Chi-2, RFE, LR, RF, LightGBM

Feature	Pearson	Chi-2	RFE	LR	RF	LightGBM	Total
Resting-blood-pressure	True	True	True	True	True	True	6
St-slope-flat	True	True	True	True	False	True	5
Max-heart-rate-achieved	True	True	True	False	True	True	5
Exercise-induced-angina	True	True	True	False	True	True	5
Cholesterol	True	False	True	True	True	True	5
Age	True	True	True	False	True	True	5
St-slope-upsloping	True	True	True	False	True	False	4
Sex-male	True	True	True	True	False	False	4
Chest-pain-type-non-anginal pain	True	True	True	True	False	False	4
Chest-pain-type-atypical angina	True	True	True	True	False	False	4
St-depression	False	False	False	False	True	True	2

Table 6 Comparison of different models after feature selection

Model	Accuracy	Precision	Sensitivity	Specificity	F1 Score	ROC	Log Loss	MCC
CardiaTics	0.933054	0.919118	0.961538	0.720588	0.939850	0.93458	2.313261	0.813662
RF	0.897872	0.872180	0.943089	0.848214	0.906250	0.895652	3.527422	0.797405
MLP	0.800000	0.775362	0.869919	0.723214	0.819923	0.796566	6.907861	0.601711
KNN	0.791489	0.776119	0.845528	0.732143	0.809339	0.788836	7.201804	0.582813
EXtra	0.893617	0.882812	0.918699	0.866071	0.900398	0.892385	3.674389	0.787059
tree classifier								
XGB	0.855319	0.856000	0.869919	0.839286	0.862903	0.854602	4.997161	0.709875
SVC	0.817021	0.794118	0.878049	0.750000	0.833977	0.814024	6.319956	0.635284
SGD	0.782979	0.750000	0.878049	0.678571	0.808989	0.778310	7.495772	0.570714
Ada-boost	0.825532	0.810606	0.869919	0.776786	0.839216	0.823352	6.025999	0.650971
CART	0.872340	0.884298	0.869919	0.875000	0.877049	0.872459	4.409253	0.744433
GBM	0.834043	0.818182	0.878049	0.785714	0.847059	0.831882	5.732049	0.668142

**Fig. 5** The before and after feature engineering confusion matrix of CardiaTics

CardiaTics also improved in all categories (marked in bold). Specifically, accuracy, precision, sensitivity, F1-score, ROC, and MCC show increases of 3.94% (93.3%), 4.8% (91.9%), 3.3% (96.1%), 3% (93%), 0.043 (0.934), and 0.03 (0.81), respectively. RF, ET, and CART also exhibit comparable performance with accuracies of 89.7%, 89.3%, and 87.2%, respectively. In contrast, the lowest-performing model is SGD, achieving an accuracy of 78.2% only. Fig. 5(a) shows the confusion matrix of CardiaTics model over raw data post feature engineering accurately predicting 215 of 240 instances. Fig. 5(b), on the other hand, shows about 4% improvement of CardiaTics post feature engineering by accurately predicting 225 of 240 instances. Note that all models show slight improvement

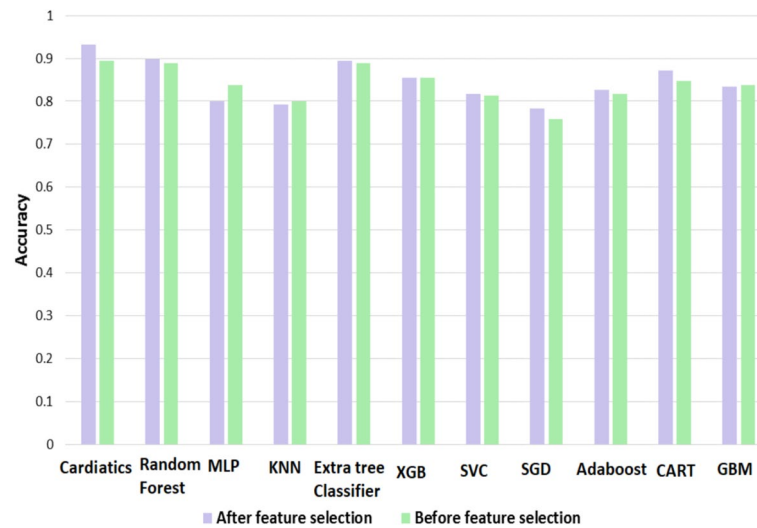


Fig. 6 Accuracy comparison different models

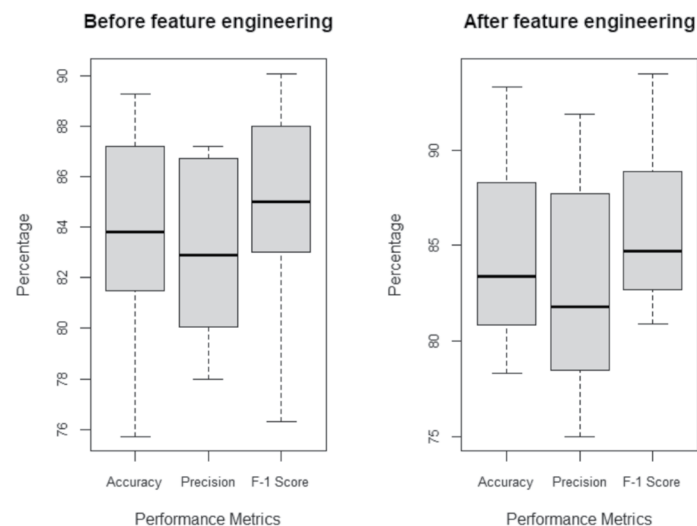


Fig. 7 Comparison of performance metrics before and after feature engineering

in performance post feature engineering as expected, except MLP and KNN. Finally, as shown in Table 6, CardiaTics improved its accuracy from 89.3% in pre feature engineering stage to 93.3% after feature engineering, outperforming all individual classifiers as compared in Fig. 6.

Statistical analysis

In this section, we conduct a thorough statistical analysis of the proposed model on performance metrics (using a cross-validation scheme) and significant nonparametric tests (using the co Friedman test and the posthoc Nemenyi test to compare models). We conducted Friedman and Nemenyi statistical analyses to evaluate how feature selection impacts model performance. These tests allowed us to identify the most predictive variables and assess their contribution to the model's accuracy and generalizability. From the boxplots figure 7 it is clear that the performance metrics don't follow the normal distribution. Hence, we introduce Friedman test to determine significance of proposed model.

Table 7 Friedman test before feature engineering

χ^2	df	p-value
11.455	2	0.003256

Table 8 Friedman test after feature engineering

χ^2	df	p-value
15.273	2	0.0004826

Table 9 Friedman effect size before feature engineering

y	n	Effsize	Method	Magnitude
Score	11	0.521	Kendall	W large

Table 10 Friedman effect size after feature engineering

y	n	Effsize	Method	Magnitude
Score	11	0.694	Kendall	W large

Table 11 Nemenyi test before feature engineering

	Accuracy	Precision
Precision	0.798	-
F-1 Score	0.028	0.004

Table 12 Nemenyi test after feature engineering

	Accuracy	Precision
Precision	0.40693	-
F-1 Score	0.02835	0.00037

From Table 7 and Table 8 we notice that in both cases the p-value < 0.05 which illustrates the significance level. Moreover, we observed that the significance of the model after feature engineering is higher than before feature engineering. Furthermore, we investigate the Friedman effect size using Kendall method where the effect size are 0.521 (Table 9) and 0.694 (Table 10) respectively for before and after feature engineering of the proposed model. In addition, we investigate Nemenyi test to identify which specific groups differ significantly from each other. From Table 11 and Table 12 we come to a decision that the precision rate significantly differs from F-1 score specially after feature engineering.

Explainability in cardiaTics

We measure explainability of predictions in CardiaTics, using ELI5 and SHAP library for global explainability that centers around determining the influence of each feature on the prediction for improved insight. This insight into feature importance enhances interpretability, providing a clearer understanding of how the model generates predictions and aiding in the identification of the most influential variables for heart disease prediction.

Global Explainability with ELI5

The ELI5 tool is used to identify the most important features through permutation importance. It facilitates extracting and visualizing features that make significant contributions to the global comprehension of the system, thereby offering insights into their broader impact. Table 13 lists features and their weights for visualization. In Table 13, in each row, the first column represents a feature, and the Weight column shows the importance of that feature in predicting the outcome, along with the variability represented by the $\pm$ values. The number after the $\pm$ symbol indicates the variation in performance across different shuffles. For example, the weight value 0.0787 ± 0.0382 for the feature *resting_blood_pressure* represents that this feature have the weight value 0.0787 for the prediction, and this weight may vary for different shuffles within the boundary of ± 0.0382 .

Features listed higher in the list are considered more important, while those at the bottom are least significant. In this example, the top five most important features are *thalassemia_reversible defect*, *resting_blood_pressure*, *thalassemia_fixed defect*, *chest_pain_type_atypical angina*, and *st_slope_flat*.

Global explainability with SHAP

The SHAP library allows for a thorough analysis of the model on a global scale, providing valuable insights into interpreting the AI system. This interpretation involves assessing the importance of features in predicting input data, as depicted in Figs. 8 and 10. On the x-axis of these figures, features are ranked by their significance, with the most influential ones listed at the top.

Fig. 8 shows the average impact of the selected features on the proposed model. The horizontal axis shows the average impact value, while the vertical axis displays the feature names ordered from the highest to lowest impact. Notably, *st_slope_upsloping* exhibits the highest average impact, whereas *st_slope_normal* has the lowest average impact on the model output.

Table 13 Feature importance of models' predictions

Feature	Weight
Resting_blood_pressure	0.0787 ± 0.0382
Thalassemia_fixed defect	0.0361 ± 0.0321
Thalassemia_reversible defect	0.0328 ± 0.0293
St_slope_flat	0.0328 ± 0.0359
Chest_pain_type_atypical angina	0.0328 ± 0.0415
Rest_ecg_left ventricular hypertrophy	0.0262 ± 0.0161
Chest_pain_type_non_anginal pain	0.0230 ± 0.0986
Chest_pain_type_typical angina	0.0197 ± 0.0245
Age	0.0197 ± 0.0245
Fasting_blood_sugar_lower than 120mg/ml	0.0164 ± 0.0000
Num_major_vessels	0.0164 ± 0.0293
Sex_male	0.0131 ± 0.0601
Max_heart_rate_achieved	0.0098 ± 0.0161
St_slope_upsloping	0.0098 ± 0.0262
Cholesterol	0.0033 ± 0.0131
Thalassemia_normal	0 ± 0.0000
Rest_ecg_normal	0 ± 0.0000
Exercise_induced_angina_yes	0 ± 0.0000
St_depression	0 ± 0.0000

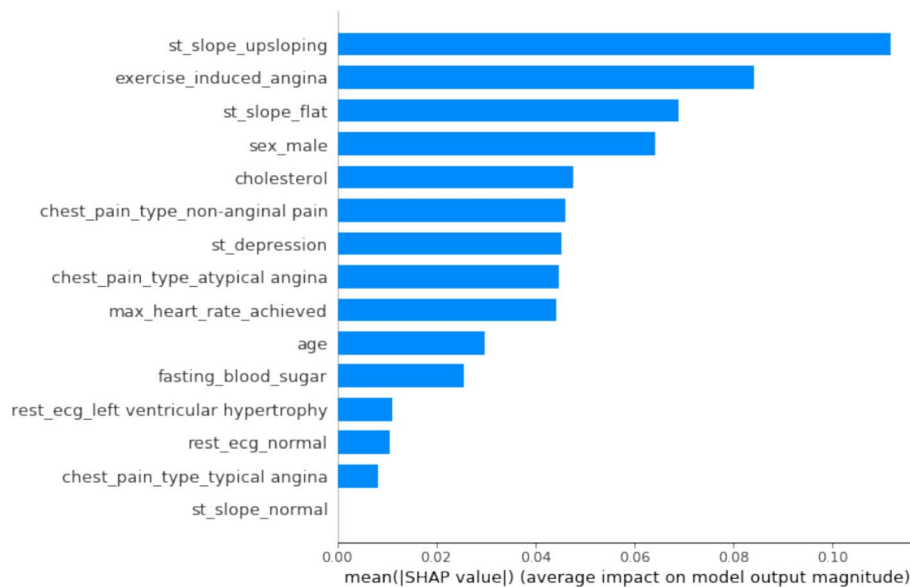


Fig. 8 This bar plot displays the mean absolute SHAP values for each feature, representing their average contribution to the model's predictions across the entire dataset. Features such as '*st_slope_upsloping*', '*exercise_induced_angina*', and '*st_slope_flat*' exhibit the highest overall influence on the model output. The x-axis quantifies the average magnitude of each feature's impact, offering insights into which variables the model relies on most when predicting heart disease risk

In Fig. 10, the 15 most important features significantly impacting the prognostic method are visually represented. The x-axis depicts SHAP values, and the y-axis displays the feature list ordered by importance from highest to lowest. The color spectrum corresponds to the original values, where blue signifies the lowest values and red signifies the highest. Note that the high values of *exercise_induced_angina*, *st_slope_upsloping*, *st_slope_flat*, and *sex_male* contribute strongly positively to the prediction, while low values have a considerable positive impact. Conversely, the *max_heart_rate_achieved* variable exhibits a highly negative contribution when its values are high and a moderately positive contribution for low values. Features like *st_slope_normal* demonstrate minimal contribution to the prediction, regardless of their values being high or low. Features identified as common by both RFE and SHAP include *st_slope_upsloping*, *exercise_induced_angina*, *st_slope_flat*, *sex_male*, *cholesterol*, *chest_pain_type_non-anginal pain*, *st_depression*, *chest_pain_type_atypical angina*, *max_heart_rate_achieved*, and *age*.

Local interpretability using SHAP

While global explainability underscores the intricate details of AI systems, local explainability, on the other hand, reveals how predictions alter with changes in specific features. This fosters trust in individual predictions and the system as a whole. Moreover, features that are significant on a global scale might not carry the same weight locally and vice versa. For local interpretation, SHAP is a commonly used tool for local interpretation, offering a force plot that visually represents the impact of each feature on the prediction model for an individual.

In Figure 9, local feature importance is illustrated using SHAP values, offering insights into individual predictions. The visualization highlights higher feature values in red (heart disease) and lower values in blue (no disease), with varying magnitudes. In this



Fig. 9 This SHAP force plot visualizes how individual features contribute to the model's prediction of heart disease risk for a specific patient. The base value represents the model's average prediction (0.5483), and the final predicted probability for this individual is 0.76. Features such as '*max_heart_rate_achieved = 163*' and '*num_major_vessels = 0*' pushed the prediction higher (toward risk), while '*thalassemia_fixed_defect = 0*' contributed to lowering it

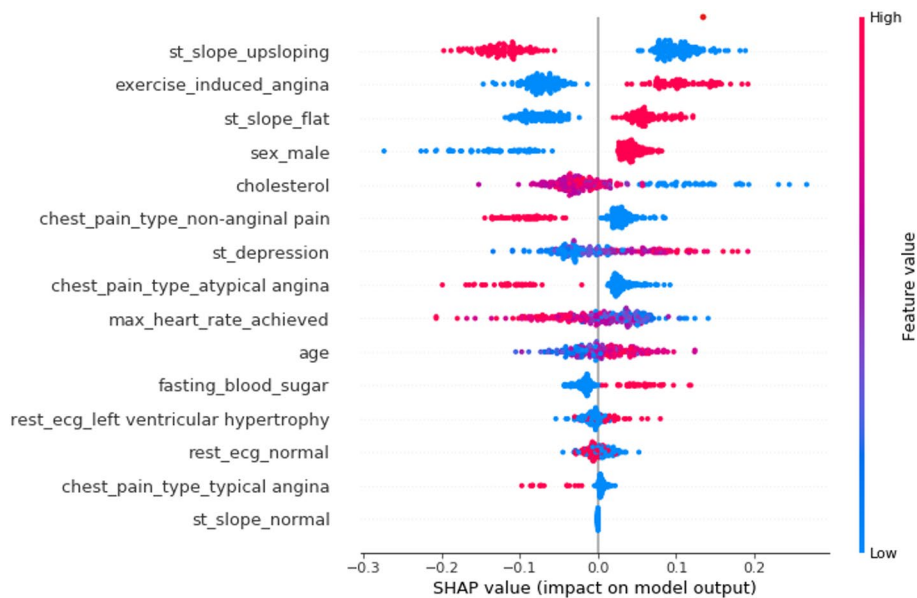


Fig. 10 This SHAP summary plot shows how each feature contributes to the model's predictions across the dataset. Each point represents a SHAP value for an individual sample, indicating whether a feature increases or decreases the predicted heart disease risk. Features such as '*st_slope_upsloping*', '*exercise_induced_angina*', and '*st_slope_flat*' have strong influence, with color representing the feature's actual value (red = high, blue = low). The plot captures both the magnitude and direction of each feature's effect, highlighting their overall importance in the model

figure, a local explanation is provided for predictions generated by a black box model. The visualization associates features such as major vessel, maximum heart rate achieved, and st_depression with the positive class, whereas thalassemia defect is correlated with the negative class. These features interact with each other, ultimately determining the highlighted prediction value for the specific observation.

Partial Dependence Plot (PDP): A PDP illustrates how changes in a single feature influence the model's predicted outcome while averaging out the effects of all other features, thereby revealing the direction and strength of that feature's contribution to the prediction.

Figure 11 and Figure 12 present SHAP-based partial dependence plots for ST-segment slope features, illustrating their marginal effects on heart disease risk prediction. The PDP shows (Figure 11) a negative contribution to the predicted heart disease risk when the upsloping ST segment is present. As the feature transitions from absence (0) to presence (1), the SHAP value decreases, indicating that an upsloping ST slope reduces the model's predicted risk. This behavior is clinically consistent, as an upsloping ST segment

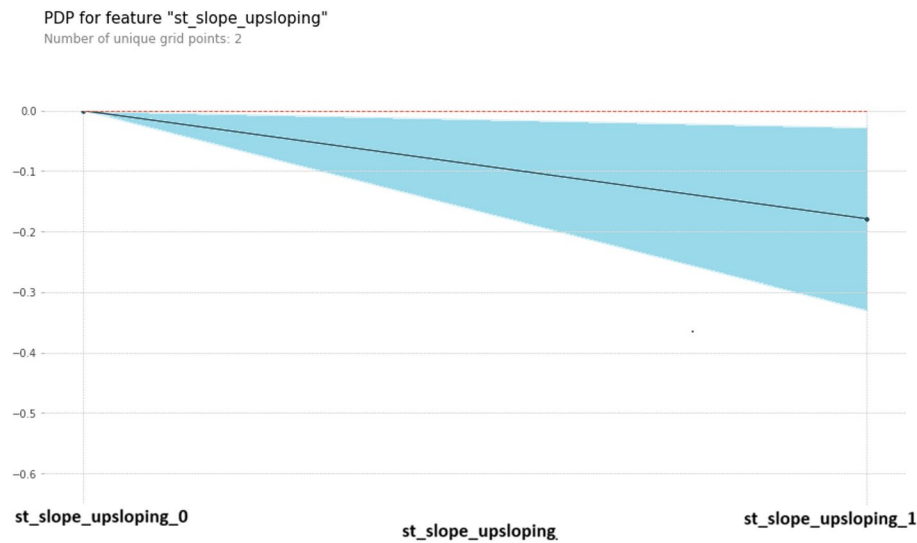


Fig. 11 SHAP partial dependence plot for *st_slope_upsloping*, showing a negative contribution to heart disease risk when the feature is present.

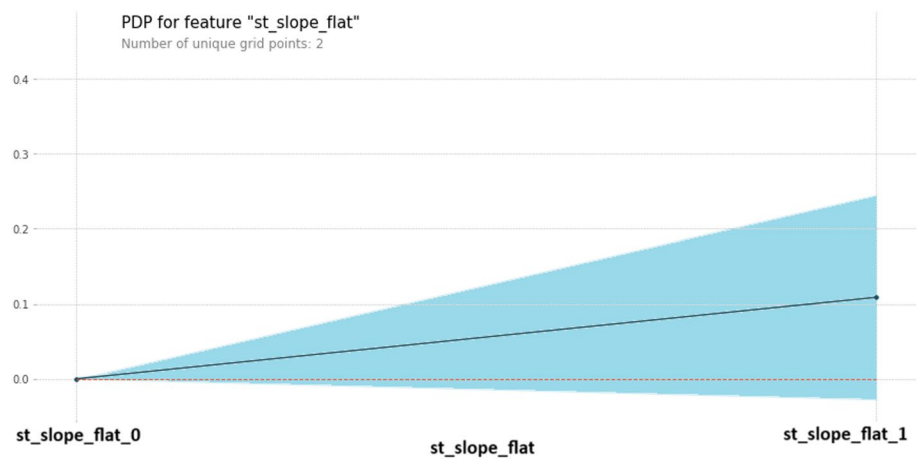


Fig. 12 SHAP partial dependence plot for *st_slope_flat*, indicating a positive contribution to heart disease risk when the feature is present.

is generally associated with a lower likelihood of ischemic abnormalities. In contrast, the PDP for flat ST slope exhibits (Figure 12) a positive contribution to the predicted risk. Moving from 0 to 1 result in an increase in SHAP values, suggesting that the presence of a flat ST segment elevates the model's predicted probability of heart disease. This aligns with clinical understanding, where a flat ST segment is often considered a pathological indicator.

SHAP provides individualized and model-agnostic feature attributions, enabling clinicians to understand not only which features influence a model's predictions but also how they do so for each patient. This level of transparency supports clinical decision-making by identifying key risk factors, uncovering unexpected model behavior, and enhancing trust in AI-assisted diagnostics. Moreover, SHAP visualizations can guide targeted interventions by highlighting the most influential variables specific to a patient's profile.

Table 14 Performance Comparison with other existing models

Reference	Methods	Dataset	FST	no. of Matrices	ACC. (%)	SEN. (%)	XAI
[36]	SVM, KNN, MLP	UCI HDD	ANOVA, EDA, PCC	4	72	80.03	NO
[27]	ANN, FNN	UCI HDD	-	2	84.2	80.3	NO
[8]	SVM, KNN, DT, Extra Tree, Bagging	UKBiobank ECG	SMOTE	4	92	-	NO
[9]	XGBoost classifier	UCI's Arrhythmia	Undersampling	4	87	88.55	NO
[12]	DT, RF, SVM, KN, LMT	University College Dublin (UCD)	Undersampling	3	91	93.83	NO
[28]	RF, DT	UCI HDD	SMOTE	4	91	82.64	NO
[17]	Deep Learning	UCI HDD	-	4	92	94.20	NO
[18]	RNN+LSTM	synthetic data	Relational approach	4	91	93.10	NO
[19]	RF	UCI HDD	Embedded approach	4	84	83	NO
[20]	CatBoost	Kaggle	Pearson's correlation	4	94	-	NO
[21]	XGBoost	UCI HDD	-	2	91.8	85.71	NO
[46]	Ensemble	ILPD	RFE	2	93.2	-	NO
CardiaTics	Ensembling	UCI HDD	Pearson's correlation, Chi-2, RFE	8	93.3	96.15	Yes

Comparison with existing works

Table 14 compares CardiaTics classification method with previous investigations. It presents a detailed comparison of various machine learning models in terms of their feature selection techniques (FST), accuracy (ACC), and explainability (XAI). Khdairet et al. (2021) explored the use of SVM, KNN, and MLP in combination with ANOVA, EDA, and PCC, achieving an accuracy of 72% across four matrices, although they lacked explainability. Liu et al. (2012) focused on SVM without specifying a feature selection method, resulting in an accuracy of 78.8% over three matrices. Kahramanli and Allah-verdi (2008) applied ANN and FNN, achieving 84.2% accuracy over two matrices, also without explainability.

Ketu et al. (2022) utilized a range of models, including SVM, KNN, DT, Extra Tree, and Bagging, along with SMOTE across four matrices, and reported a high accuracy of 92%. Rezaei et al. (2021) employed XGBoost with undersampling and achieved 87% accuracy over four matrices. Plati et al. (2021) implemented various models such as DT, RF, SVM, KNN, and LMT with undersampling, reaching an accuracy of 91% over three matrices. Similarly, Minou et al. (2020) used RF and DT with SMOTE and reported an accuracy of 91% for four matrices.

Barhoom et al. (2022) achieved a high accuracy of 92% using a deep learning approach across four matrices. Bhavakar et al. (2022) applied a hybrid RNN + LSTM method with a relational approach, achieving an accuracy of 91%. Ansarullah et al. (2022) utilized RF with an embedded approach, resulting in an accuracy of 84%. Wei et al. (2023) reported the highest accuracy of 94% using CatBoost with Pearson's correlation over four matrices. Budholiya et al. (2022) achieved 91.8% accuracy with XGBoost over two matrices.

The CardiaTics model stands out by employing an ensembling approach that combines Pearson's correlation, Chi-2, and RFE feature selection techniques. This method achieved an impressive accuracy of 93.3% across eight matrices and was notable for incorporating explainability (XAI), making it comparable to other heart disease prediction models.

Our optimized model outperforms (marked in bold) in heart disease detection, showing superior accuracy in eight total set of metrics – Accuracy, Precision, Sensitivity, Specificity, F1-Score, ROC, LogLoss, and MCC. From the table, we also observe that only CardiaTics incorporated model explainability as a feature of the system.

Limitations and future plans

Although CardiaTics demonstrates strong predictive performance and interpretability on the benchmark dataset, several limitations should be acknowledged. First, an experimental evaluation was conducted on a UCI-based heart disease dataset comprising a limited number of samples and features. Although this dataset is widely used for benchmarking and allows for fair comparison with existing studies, it may not fully capture the variability present in real-world clinical environments. Consequently, the generalizability of the proposed framework to external institutional datasets requires further validation. Second, CardiaTics employs a stacked ensemble architecture that integrates multiple heterogeneous base and meta learners. Although this design improves predictive robustness and performance, it introduces additional computational overhead compared to single-model approaches. However, this trade-off is considered acceptable given the clinical context, where reliability and interpretability are often prioritized over marginal increases in computational costs.

In terms of future work, the framework can be extended by evaluating CardiaTics using external and multi-institutional datasets to assess its robustness across diverse populations and healthcare settings. Model calibration, retraining, and fine-tuning strategies may be explored to address potential domain shifts arising from differences in data acquisition protocols or patient demographics. Furthermore, future studies may investigate domain adaptation techniques and incremental learning to support broader deployment scenarios. We also plan to develop a cloud-based CardiaTics application based on the proposed framework. This platform will enable the collection of diverse real-world clinical data from practitioners, facilitating continuous model refinement and validation. By incorporating newly acquired data, the system aims to improve predictive performance, robustness, and generalizability, ultimately supporting more accurate heart disease assessments and improving patient outcomes.

Implementation challenges and advantages

The CardiaTics image analysis system is hosted on the University of Idaho's high-performance computing infrastructure at <http://dblab.nkn.uidaho.edu/cardiatics/> and operates in full compliance with HIPAA regulations. All images stored in the system are stripped of patient identifiers and are submitted voluntarily for research purposes only. Each submission undergoes a manual compliance review and is treated as a research contribution. Image ownership remains with the depositor, and the data is not publicly accessible or shared with other users. Internally, images are assigned unique identifiers specific to CardiaTics that are not linked to patient information, ensuring privacy. These images are used exclusively to improve the CardiaTics system and are never employed for commercial purposes.

CardiaTics is also evaluating a HIPAA-compliant image sharing protocol currently being tested as part of the PerHL (Personal Health Library) initiative [47, 48]. This protocol is designed to facilitate secure collaboration between clinicians and researchers,

enhancing diagnostic accuracy and promoting knowledge sharing. Importantly, CardiaTics is not a repository for medical records and cannot be connected to health information systems. Its focus is strictly on cardiological image analysis using advanced machine learning models not available through existing platforms. The system emphasizes ethical sharing of analytical tools – such as ML models – rather than data or images themselves.

Conclusion

In this paper, introduced a stacked machine learning ensemble classifier, called CardiaTics, that used several machine learning techniques to provide a more precise and reliable model for detecting the presence of a heart disease. We choose the top 10 models for level 0, or base level, of the stacked ensemble based on how well each baseline model performed in cross-validation accuracy. Then we fed the test data from each of the core classifiers into an MLP algorithm to select the optimal ensemble of the group of classifiers. We also used Pearson correlation, Chi-2, and RFE techniques for feature selection using feature engineering. The stacked classifier model outperformed each individual classifier, with an accuracy of 89.3% on the raw data and 93.3% after feature selection. The results underscore the potential of this stacked model as a valuable tool for healthcare practitioners in improving their diagnostic capabilities. Furthermore, the interpretability of the model's decision-making process provides valuable insights, aiding clinicians an understanding of the factors contributing to each prediction.

As we move forward, it is important to continue refining and expanding the model by incorporating additional data sources to enrich medical knowledge, while exploring domain adaptation strategies to address potential distribution shifts in future work. We have plans to develop a CardiaTics cloud application based on this approach so that more novel data could be collected from practitioners so that the model can be improved, and play a role in understanding heart disease through improved predictions for more impactful patient outcomes.

Acknowledgements

This research was supported in part by the National Institutes of Health IDeA grant P20GM103408, and the National Science Foundation CSSI grant OAC 2410668.

Author contributions

P.G: Conceptualization of this study, Data curation, Writing - Original draft preparation, Methodology, Investigation, Writing- Review & Editing H.J: Investigation, Supervision, Writing – review & editing

Funding

This research was supported in part by the National Institutes of Health IDeA grant P20GM103408, and the National Science Foundation CSSI grant OAC 2410668.

Data availability

No datasets were generated or analysed during the current study.

Materials availability

Not applicable.

Code availability

The data and code will be made available upon request.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 11 November 2024 / Accepted: 17 February 2026

Published online: 04 March 2026

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